Skeleton Tutorial Template

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Chapter 1

Opening page

Basically a standard Bookdown template with a few tweaks. New chapters need to be in separate '.Rmd' files, where each file starts with a chapter heading as seen here. In order to use the task and solution blocks in LaTeX, you must input the order of the files into the `_bookdown.yml` file, and the first file must be called `index.Rmd` e.g.

```
rmd_files:
    html: ['index.Rmd', 'ch1.Rmd']
    latex: ['index.Rmd', 'ch1.Rmd', 'ch_appendix.Rmd']
output_dir: "docs"
```

The `latex:` path above ***must*** have `'ch_appendix.Rmd'` as its last entry. This ensures that the appendix is properly formatted for the solutions to the problems.

You must have the following lines at the start of your `index.Rmd` file:

```
```{r, child = "_setup.Rmd", include = F, purl = F, cache = F}
```
```

There are a couple of useful special blocks. A `task` block, and a `solution` block. These can be used as e.g.

```
```{task}
Here is a task written in **markdown**.
```
```

which renders as:

Task 1

Here is a task written in **markdown**.

You can include chunks within the `task` chunk, but you need to use double backticks *within* the chunk, and leave carriage returns around the internal chunk e.g.

```
```{task}

``{r}
x <- 2 + 2
x
``

```
```

which renders as:

Task 2

```
x <- 2 + 2
x
```

```
## [1] 4
```

Be careful to have suitable carriage returns around e.g. `enumerate` or `itemize` environments inside the chunk also. For example:

```
```{task}
Here is a list:
1. item 1
2. item 2
```
```

will not render nicely. But

```
```{task}
Here is a list:

1. item 1
2. item 2

```
```

will:

Task 3

Here is a list:

1. item 1
2. item 2

The `solution` chunk works in the same way, and the numbers will follow the previous `task` chunk (so you can set tasks without solutions) e.g.

```
```{task}
Add 2 and 2 together
```

```{solution}

``{r}
2 + 2
``

```
```

gives:

Task 4

Add 2 and 2 together

Show: Solution on P11

1.1 Additional extensions

1.1.1 Different task and solution titles

Task and solution boxes can also be given different names using the `title` option e.g.

```
```{task, title = "Question"}
What is the meaning of life, the universe and everything?
```

```{solution, title = "Answer"}
Why 42 of course!
```
```

gives:

Question 5

What is the meaning of life, the universe and everything?

Show: Answer on P11

1.1.2 Turning tasks and solutions on and off

Sometimes you might want to hide task and/or solution boxes. This can be done with the `renderTask` and `renderSol` chunk options, which can be set globally or locally. For example:

```
```{task, title = "Question"}
Can I set a task and not show the answer?
```

```{solution, title = "Answer", renderSol = FALSE}
Indeed, though you won't see this answer unless `renderSol = TRUE`...
```
```

typesets as:

Question 6

Can I set a task and not show the answer?

1.1.3 Generic information environments

You can also set generic boxed environments containing arbitrary information.

```
```{info, title = "Some interesting titbit"}
This box contains invaluable information!
```
```

typesets as:

Show: Some interesting titbit on P13

Note that it is useful to set the `title` option here, else it defaults to `info`. You can also use this environment to simply display an alert box with information, by setting the `collapsible` argument to `FALSE` in the chunk options e.g.

```
```{info, title = "Some interesting aside", collapsible = FALSE}
Yet more valuable information - this time displayed directly!
```
```

typesets as:

> Some interesting aside
>
> Yet more valuable information - this time displayed directly!

In the PDF output, setting `collapsible = TRUE` will place the information boxes in a separate Appendix, with links in the main document. You can again hide the `info` boxes by setting `renderInfo = FALSE` in the chunk options.

1.1.4 Tabbed boxed environments

Originally developed to put base R and `tidyverse` solutions side-by-side, using a `multCode = T` option to the solution box. Here the two tabs are separated by four consecutive hashes: `####`, and the `titles` option gives the tab titles (these can be set globally if preferred) e.g.

```
```{task}
Filter the `iris` data by `Species == "setosa"` and find the mean `Petal.Length`.
```

```{solution, multCode = T, titles = c("Base R", "tidyverse")}

``{r}
## base R solution
mean(iris$Petal.Length[
    iris$Species == "setosa"])
``

####

``{r}
## tidyverse solution
iris %>%
    filter(Species == "setosa") %>%
    select(Petal.Length) %>%
    summarise(mean = mean(Petal.Length))
``

```
```

will typeset to:

Task 7

Filter the `iris` data by `Species == "setosa"` and find the mean `Petal.Length`.

Show: Solution on P11

Note that there is also a `multCode` chunk that does not link to task and solution boxes e.g.

```
```{multCode}

Two options:

* Option 1

####

Two options:

* Option 2

```
```

will typeset to:

Option 1

Two options:

- Option 1

Option 2

Two options:

- Option 2

The `titles` option can be set as before.

Appendix A

Answers

Solution 4

```r
2 + 2
```

```
## [1] 4
```

Return to task on P7

Answer 5

Why 42 of course!

Return to task on P8

Solution 7

Base R

```r
## base R solution
mean(iris$Petal.Length[
    iris$Species == "setosa"])
```

```
## [1] 1.462
```

tidyverse

```r
## tidyverse solution
iris %>%
    filter(Species == "setosa") %>%
    select(Petal.Length) %>%
    summarise(mean = mean(Petal.Length))
```

```
##    mean
## 1 1.462
```

Return to task on P10

Appendix B

Additional information

Some interesting titbit

This box contains invaluable information!

Return to P8